Drone model/year	Developer/country	Key features and capabilities	Primary role/applications	Techn signif
				stealth UCAV
X-47B (2012)	Northrop Grumman/United States	Carrier-based UCAV, autonomous takeoff and landing, precision navigation	Naval reconnaissance and strike operations	First autonc operat aircra carrie demor advan autonc
Wing Loong I (2015)	Aviation Industry Corporation of China (AVIC)/China	MALE UAV, EO/IR sensors, SATCOM, armed with guided munitions	ISR, precision strike, export platform	Offere effecti export UAV to defens marke

4.6 Challenges and Limitations of Third-generation Drones

Despite their remarkable advancements in endurance, autonomy, and mission versatility, third-generation drones faced several technological, operational, and strategic challenges that limited their overall effectiveness and scalability. These limitations played a critical role in shaping the direction of fourth-generation UAV development, emphasizing enhanced autonomy, stealth, and integration with artificial intelligence.